

GEHC 5338648
MR MDP Battery & Charger Kit Installation Instructions
FOR PANELS 243B10431, 243B10435, 243B10438, 243B10449, & 243B10453

Notice:

The person performing this work must be qualified to work on high voltage electrical equipment.

UL Recognized E61758

Purpose

The purpose of this kit is to replace the battery and battery charger with a new battery and power supply in the Main Disconnect Panel (MDP). There are 5 MDPs that are used on MR systems. MR MDPs are manufactured by GE Consumer and Industrial:

GE C&I Catalog #	GE Healthcare Catalog #	GE C&I Diagram #
243B10431	G0000AL	55-686092 or 55-216700
243B10435	M3088TH	55-686153 or 55-216870
243B10438	E4503K	55-686175 or 55-216937
243B10449	n/a	55-686232 or 55-217169
243B10453	M3088TM	55-788127

To identify your MDP, you will need to open the panel door and look for one of the numbers listed above. The number will generally be located on the schematic or component layout drawing located on the inside of the door or on an individual label. Mark the catalog number on the panel you are working on in the table above for future reference.

Before opening the door, turn off all incoming power sources and apply correct LOTO (Lock Out Tag Out) procedures and equipment as required locally. Follow PERSONAL PROTECTIVE EQUIPMENT POLICES of the installer's company and of the site owner.

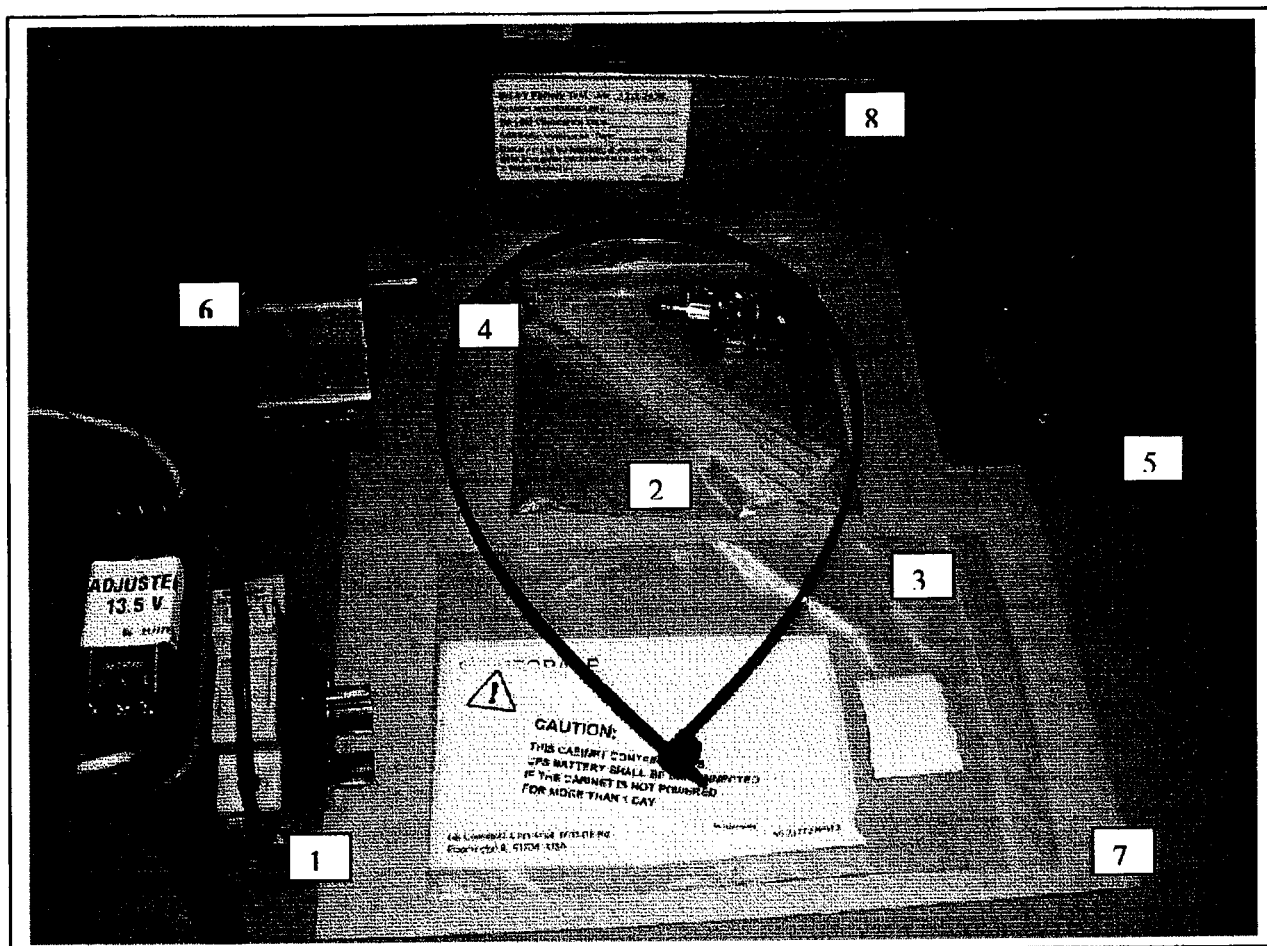
If you are unable to locate a GE C&I 243B10... part number or are having difficulty identifying your MDP, contact GEXPRO Milwaukee at 414-527-6600.

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Parts Included



1. Power supply assembly
2. Bag containing terminals, screws and wire ties
3. Bag containing labels
4. Cable strap to secure battery
5. Mounting plate
6. Z-Bracket
7. Envelope with diagrams and installation instructions
8. Battery

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LIST OF MATERIAL FOR 5338648

PART NUMBER	DESCRIPTION	QUAN	COMMENTS
3007N36P013	FUSE BLOCK	1	
3006N35P020	2 AMP FUSE LP-CC-2	1	
3011N10P003	DC POWER SUPPLY ML15.121	1	
4002N86P013	TERMINAL BOARD END STOP	1	
4002N86P112	TERMINAL BOARD MARKER "G"	2	
4002N86P027	TB SEGMENT - GROUND	1	
55-217729P002	DIN RAIL 3 1/4"	1	
N50KC41JF	#14 DK BLUE WIRE	34"	
4000N10P046	POWER CORD	1	
4000N29P002	CABLE TIE (14 3/4")	1	
55-216961P006	GREEN/YELLOW #14 WIRE	60"	
N50KC41JT	#14 WHITE/BLUE WIRE	28"	
4000N29P0011	CABLE TIE 5 1/4"	6	
4000N15P097	PIGGY BACK TERMINAL	2	
4000N15P098	TERMINAL 16-14 AWG 0.250	4	
N168P1606B6	SCREW #10 X 3/8"	5	
N722EP16006B6	SELF TAPPING SCREWS	5	
4001N43P001	BATTERY PS-1270 F2	1	
55-217727P006	TEMPERATURE LABEL	1	
4000N29P027	CABLE TIE 24" X 0.3	1	
55-217749P001	MOUNTING PLATE	1	
55-217749P002	Z-BRACKET	1	
N51P13008B6	#6 X 1/2" FLAT HEAD SCREW	3	
55-217727P007	LABEL "NON SPILLABLE BATTERY"	2	
55-217727P019	LABEL "UPGRADE"	1	
55-217727P005	LABEL "INSTALLED DATE"	1	
55-217727P016	LABEL "+"	2	
55-217727P017	LABEL "-"	2	
55-217727P025	LABEL "ADJUSTED 13.5V"	1	
55-217726P001	DRILL BIT 0.173	1	
55-217727P033	LABEL "CAUTION"	1	
55-217751	INSTALLATION GUIDE	1	
55-788161	WIRING DIAGRAM FULL SIZE	1	FOR PANEL 243B10453
55-788161	WIRING DIAGRAM -B SIZE ADHESIVE	1	FOR PANEL 243B10453
55-686564	WIRING DIAGRAM FULL SIZE	1	FOR PANEL 243B10431
55-686564	WIRING DIAGRAM -A SIZE ADHESIVE	1	FOR PANEL 243B10431
55-686565	WIRING DIAGRAM FULL SIZE	1	FOR PANEL 243B10435
55-686565	WIRING DIAGRAM -A SIZE ADHESIVE	1	FOR PANEL 243B10435
55-686566	WIRING DIAGRAM FULL SIZE	1	FOR PANEL 243B10449
55-686566	WIRING DIAGRAM -A SIZE ADHESIVE	1	FOR PANEL 243B10449
55-686570	WIRING DIAGRAM FULL SIZE	1	FOR PANEL 243B10438
55-686570	WIRING DIAGRAM -A SIZE ADHESIVE	1	FOR PANEL 243B10439
	BATTERY INSTRUCTION	1	
	POWER SUPPLY INSTRUCTION	1	

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Tools Required for Installation

Phillips Screwdriver, medium point #2
 Straight Blade screwdriver 1/8" tip
 Straight Blade Screwdriver, medium tip
 Wire cutter (#14 wire)

Wire strippers (#14 wire)
 Crimpers (#14 Insulated Terminal)
 Digital Volt Ohm Meter, 4 ½ digits

Installation Preface

Battery Check

Locate the new battery from the kit. Check the temperature sticker on the new battery and the new battery voltage. If the sticker shows a temperature of more than 120 degrees F (49C) or if the battery voltage is less than 12VDC the battery should not be used. Contact GEXPRO Milwaukee at 414-527-6600 for a replacement battery.

Power Supply Mounting

There are three mounting arrangements available for the new power supply assembly. They are:

- Receptacle mount
- Z-bracket mount
- Direct Panel mount.

Refer to the table below for the recommended mounting method.

Mounting Method	MDP Catalog #				
	243B10431	243B10435	243B10438	243B10439	243B10453
Receptacle	Not Recommended	Preferred	Preferred	Preferred	Preferred
Z Bracket	Preferred	Alternate	Alternate	Alternate	Alternate
Direct	Alternate	Alternate	Alternate	Alternate	Alternate

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If you have not already done so, disconnect all incoming power and apply correct LOTO (Lock Out Tag Out) procedures and equipment as required locally.

Follow PERSONAL PROTECTIVE EQUIPMENT POLICES of the installer's company and of the site owner.

Installation

1) Disconnect all incoming power and apply correct LOTO (Lock Out Tag Out) procedures and equipment as required locally. Follow PERSONAL PROTECTIVE EQUIPMENT POLICES of the installer's company and of the site owner.



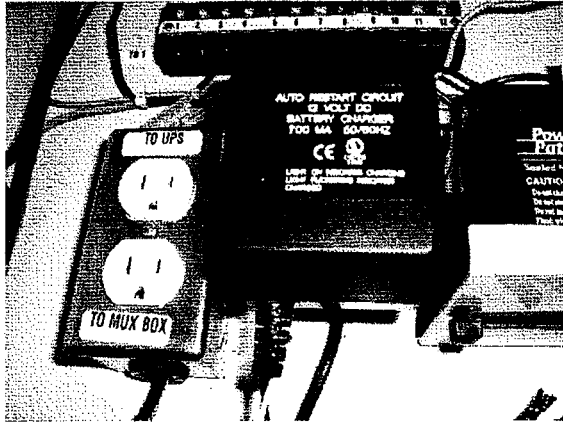
Note: View looking down from top of panel.

2) Open panel door. Verify that blue wires connected to the battery are marked with plus (+) and minus (-) labels before removing wires from battery. If not marked, mark them with the labels included in kit. Disconnect wires from battery.

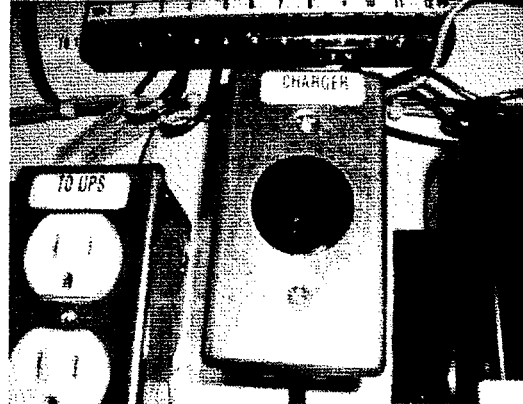
Note: It is possible that the piggyback terminals for the battery connection could be damaged when disconnecting the battery. Replacement terminals have been included in the kit, if required.

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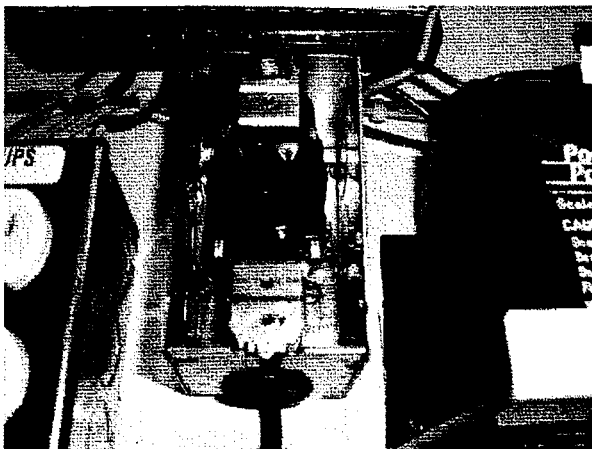
3) Cut and remove cable tie around charger



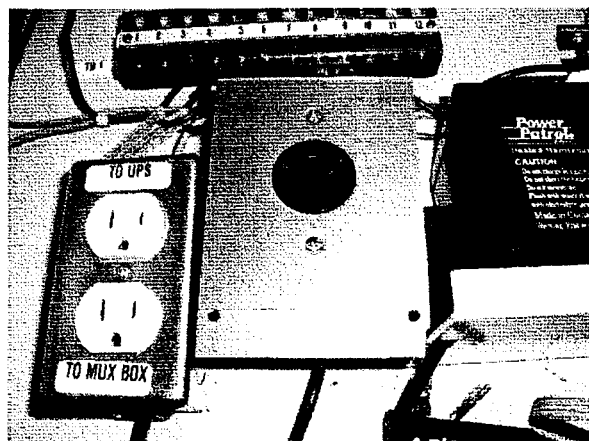
4) Remove the charger from the receptacle. Remove charger wire leads from piggyback terminals.

For Receptacle mount, go to step 5, below
For Z bracket mount, go to step 9, Page 8
For Direct mount, go to step 10, Page 9

Receptacle Mount



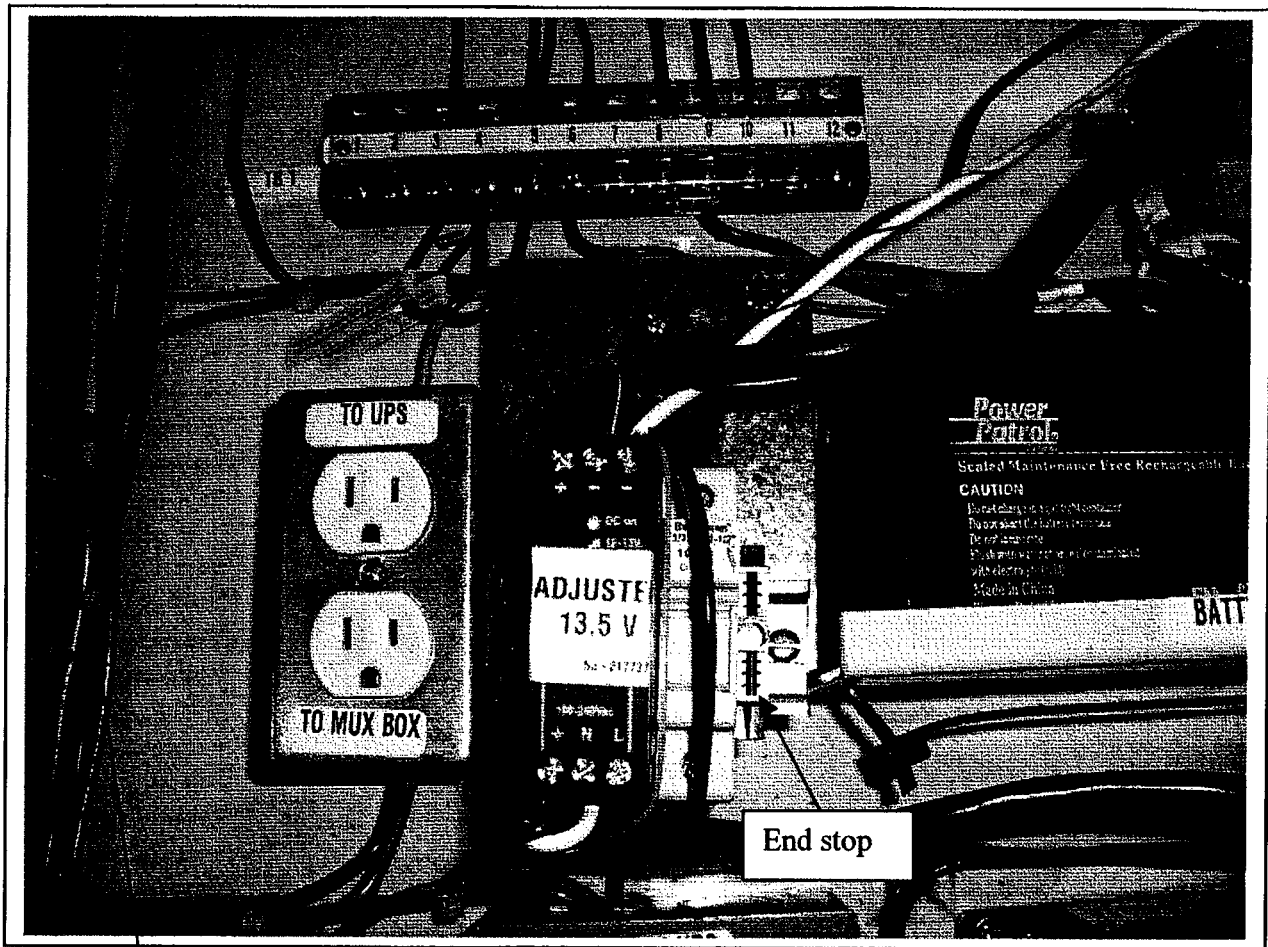
5) Remove the cover from the receptacle and discard. Save the two receptacle mounting screws.



6) Fasten the new cover/mounting plate (included in the kit) to the receptacle using the saved screws or new screws from kit. For orientation see following note

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Note: The cover plate can be positioned with the power supply din rail mounting holes above or below the receptacle. Choose the position for the least interference of panel components and cleanest lead dress.



7) Connect ground wire to ground bar on left side of enclosure. Use a wire tie from the kit to secure to wire bundle.

8) Fasten the power supply assembly to the mounting plate as shown. Secure the mounting rail to the plate with 2 screws in the tapped holes.

Receptacle Mount continue at Step 11, Page 10

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Z-Bracket Mount



Z Bracket

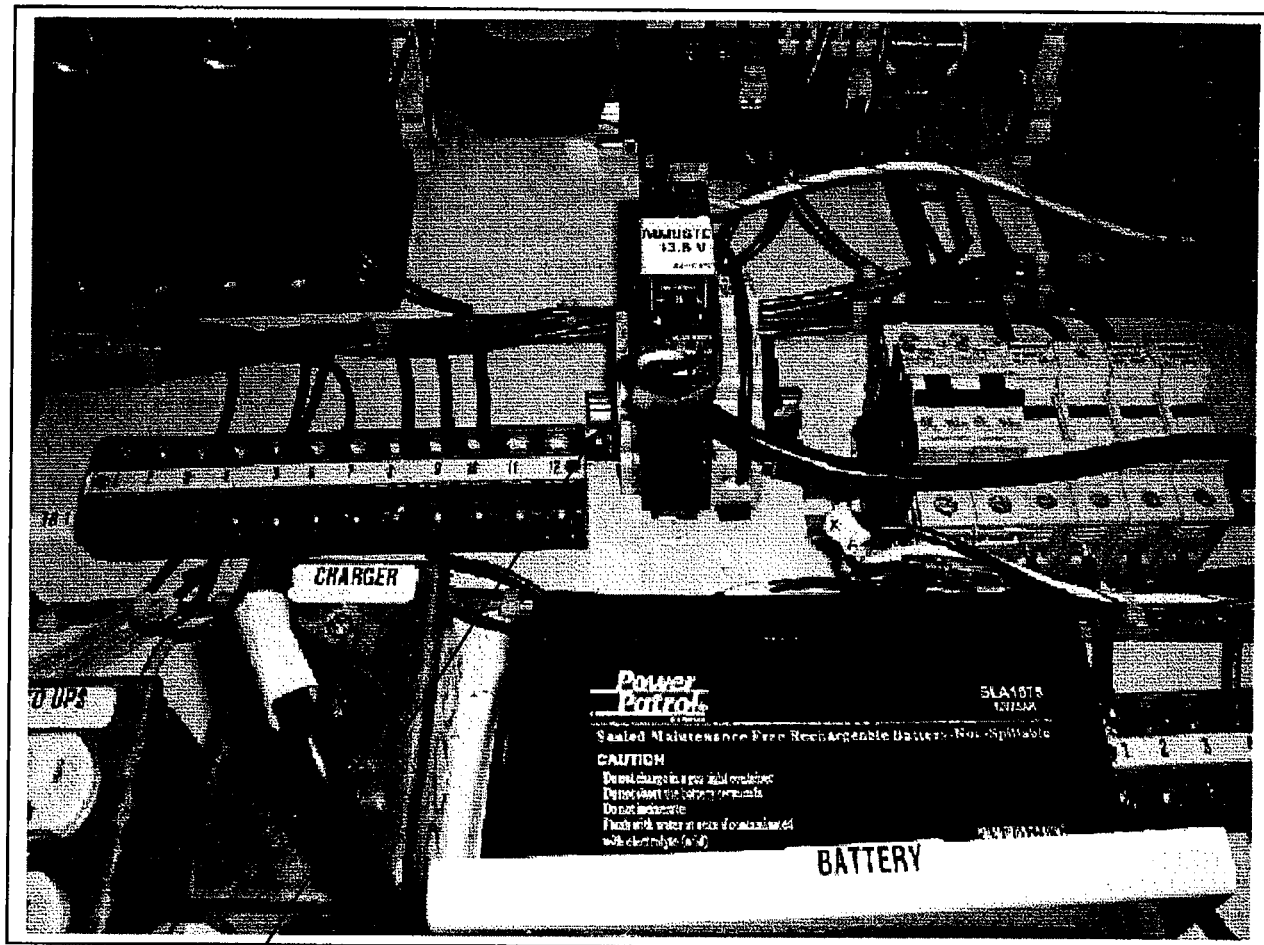
Self-tapping screws

9) Mount Z-bracket in a location where there is no interference with existing components and wiring. See photo for example. Use Z-bracket as a guide for drilling and tapping of holes into the back of the enclosure. A drill bit and self-tapping screws are provided in the kit. To fasten the assembly to the Z-bracket, mount din rail assembly as shown onto upper surface of Z-bracket.

Z-Bracket Mount continue at Step 11, Page 10

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Direct Mount



Self-tapping screws

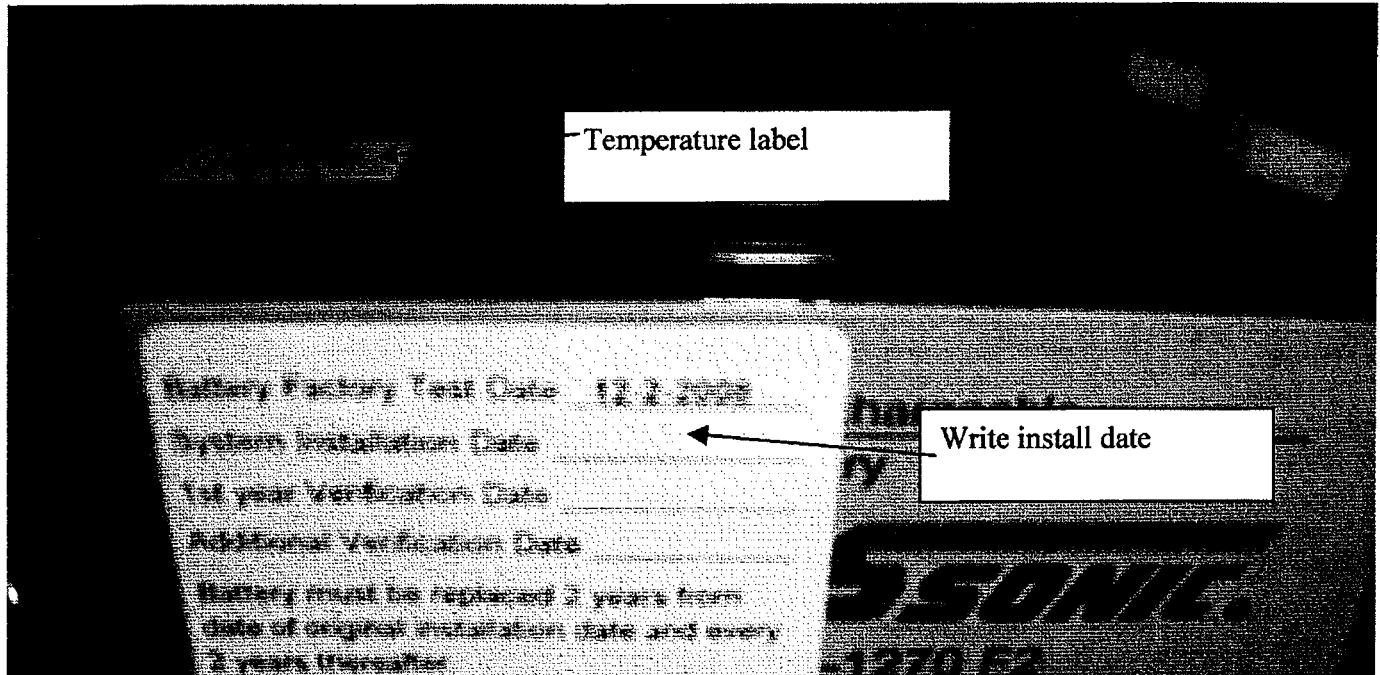
10) Mount din rail assembly in a location where there is no interference with existing components and wiring. See photo for example. Use the din rail as a guide for drilling and tapping of holes into the back of the enclosure. A drill bit and self-tapping screws are provided in the kit.

Direct Mount continue at Step 11, Next page

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Battery Replacement



- 11) Mark the installation date on the new battery.
- 12) Cut and remove the cable tie holding the battery to the bracket.
- 13) Remove the old battery in the panel and replace it with the battery included in the kit.
- 14) Remove and save the battery terminal insulators from the new battery for the return of the used battery.
- 15) A new strap, to fasten the battery in place, is included in the kit. Put the new battery in place and tie down tightly with the new strap.
- 15a) Before connecting any wires to the battery, measure and record the battery terminal voltage.

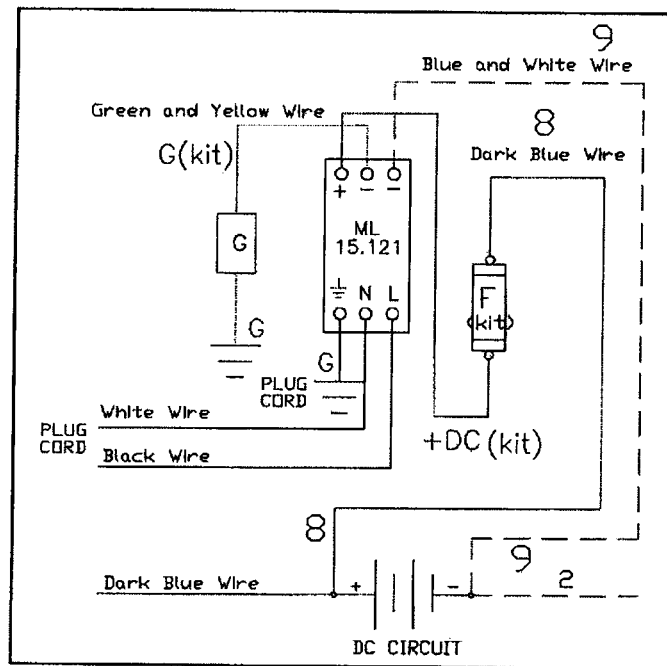
_____ Volts DC.

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16) Connect the existing #14 dark blue positive (+) wire (piggy back terminal) to the positive (red) terminal on the battery and the existing #14 blue / white negative (-) wire (piggy back terminal) to the negative (black) terminal on the battery.

17) Place the battery terminal insulators from the new battery on to the terminals of the used battery or use electrical grade tape to insulate terminals on the old battery.



18) Connect from the new power supply the #14 blue / white wire (marked negative with a [-] label), to the piggy back terminal attached to the battery negative (black) terminal.

19) Connect from the new power supply the #14 dark blue wire (marked positive with a [+] label), to the piggy back terminal attached to the battery positive (red) terminal. **Note:** The Power Supply green "DC ON" LED will turn on when the power supply is connected to the battery.

20) Plug the cord into the single outlet (the receptacle that was used for the old battery charger)

21) Use the wire ties that were included in the kit to tie wires if needed.

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Labeling

There are 5 different wiring diagrams included with this kit. The identification of the panel is the same as described in the "Purpose" section on page 1. To determine which one is needed for the panel being upgraded, determine the panel catalog number from the "GE" label inside the cover. The number will start with "243". For instance "243B10435". Then look at the "List of Material" on page 12, of this guide, to determine which diagram is associated with the panel being upgraded. (In this case the diagram number needed is 55-686565). Remove the correct full sized and reduced sized, adhesive backed diagrams from the envelope and discard the others. Additional verification may be made by making sure the diagram legend contains the panel catalog number. The chart below lists the 5 MDPs and their respective numbers.

GE C&I Catalog #	GE Healthcare Catalog #	<u>Existing</u> GE C&I Diagram #	<u>New</u> GE C&I Diagram #
243B10431	G0000AL	55-686092 or 55-216700	55-686564
243B10435	M3088TH	55-686153 or 55-216870	55-686565
243B10438	E4503K	55-686175 or 55-216937	55-686570
243B10449	n/a	55-686232 or 55-217169	55-686566
243B10453	M3088TM	55-788127	55-788161

22) Fasten the "WIRING UPGRADE" label 55-217727P019 and the adhesive backed wiring diagram inside the cover. The new diagram should cover the existing diagram inside the cover. The original full size wiring diagram should be removed from the equipment instruction package and replaced with the new full size wiring diagram.

23) Fasten the "Caution" 55-217727P033 label on the front of the panel.

24) Clean out panel of debris such as metal shavings, unused wires, etc.

25) Verify that all wiring is neatly bundled and routed along the cable bundles to prevent kinks or pinched wires.

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Testing

26) Verify that the panel main disconnect is in the OFF position.

27) Make sure all Remote Emergency Off functions or pushbuttons are set to normal ("RUN") position.

28) Verify that all connected equipment is turned off.

Note: For steps 29 and 30, the panel door needs to be opened and the panel energized. Use proper PPE including Flash Guard if required.

29) Remove the lock and lockout tag following the established lockout removal procedures. ***Confirm arc flash PPE requirements before attempting to open the energized panel. Use proper PPE clothing before attempting to test, service or troubleshoot any energized panel. Less than category II protective clothing (and hood) shall NEVER be used..*** With the panel door open, turn the panel disconnect or the external disconnect On to apply power to the panel. For MDPs that have a main disconnect, a lock-out device may have to be actuated in order to operate the disconnect with the panel door open. The green LED light on the power supply should be on. If not, recheck connections, check the fuse in the DIN rail mounted fuse holder included with the kit, and verify 120Vac at the Power Supply "L" to "N" terminals.

30) Measure the voltage at the Power Supply "+" to "-" terminals .The voltage should measure:

13.5 VDC +/-0.05VDC.

To adjust the voltage, locate the potentiometer under the "Adjusted 13.5 VDC" label. Adjust the pot using a small screwdriver. Replace the label.

Note: The correct setting of the voltage is critical for long battery life.

31) Measure the voltage at the battery terminals. This voltage should be greater than the battery voltage measured in step 15a. If not, verify 120VAC +/- 10% at the power supply L to N terminals and check the fuse in the DIN rail mounted fuse holder included with the kit.

After an overnight charge, the battery voltage should be 13.5 +0.05 / -0.15V.

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32) Deliver or ship the old battery to a battery-recycling center. A “NON SPILLABLE BATTERY” label is included in the kit and should be placed on the outside of the carton used to ship the battery.

Kit installation is complete.